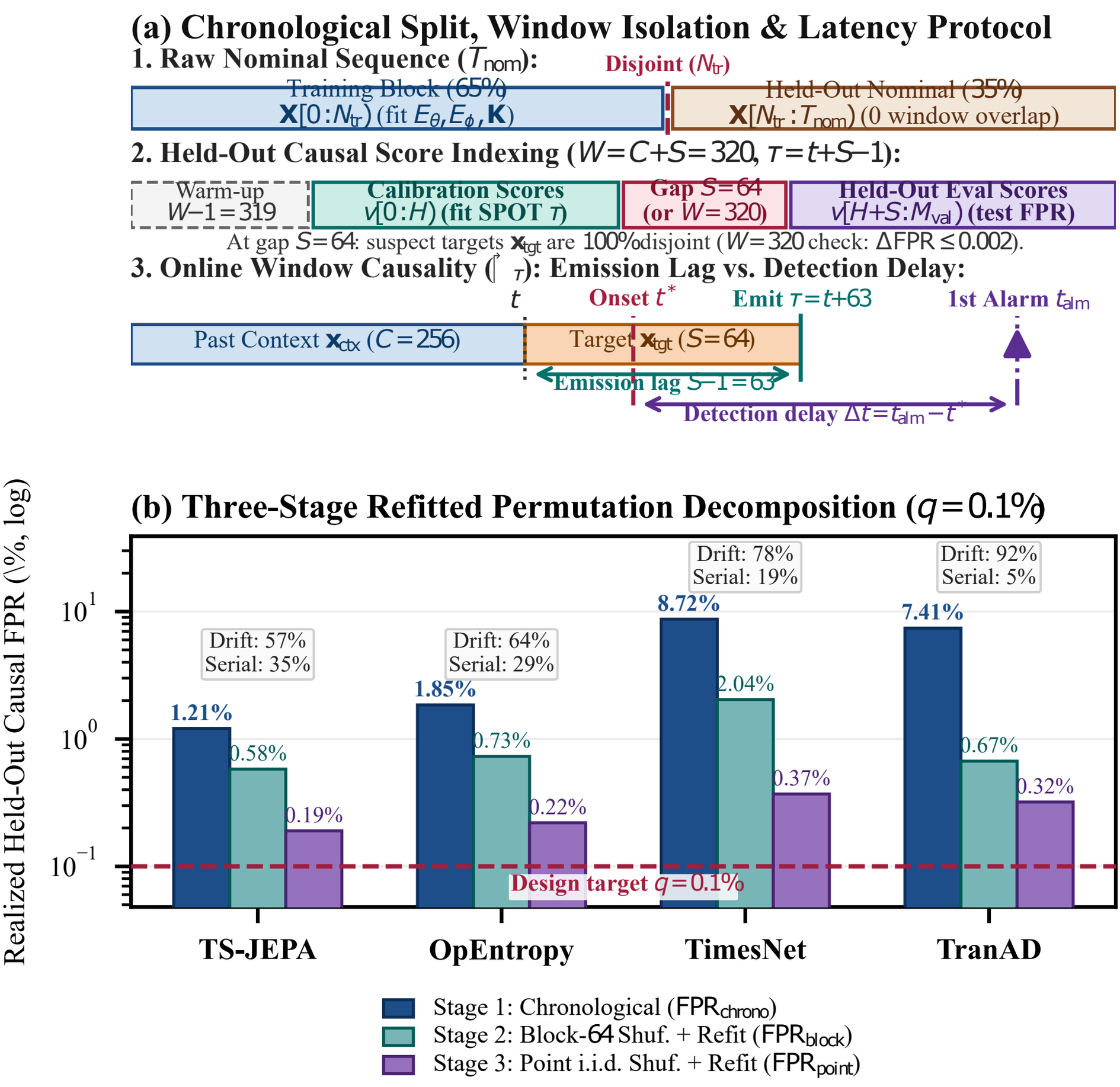

(a) Chronological Split, Window Isolation & Latency Protocol
1. Raw Nominal Sequence (T_{nom}):
Disjoint (N_{tr})
Training Block (65%)
$\mathbf{X}[0:N_{tr})$ (fit $E_\theta, E_\phi, \mathbf{K}$)
Held-Out Nominal (35%)
$\mathbf{X}[N_{tr}:T_{nom})$ (0 window overlap)
2. Held-Out Causal Score Indexing ($W=C+S=320$, $\tau=t+S-1$):
Warm-up
$W-1=319$
Calibration Scores
$v[0:H)$ (fit SPOT τ)
Gap $S=64$
(or $W=320$)
Held-Out Eval Scores
$v[H+S:M_{val})$ (test FPR)
At gap $S=64$: suspect targets $\mathbf{x}_{tgt}$ are 100% disjoint ($W=320$ check: ΔFPR ≤ 0.002).
3. Online Window Causality (τ): Emission Lag vs. Detection Delay:
t
Onset t^*
Emit $\tau=t+63$
1st Alarm t_{alm}
Past Context $\mathbf{x}_{ctx}$ ($C=256$)
Target $\mathbf{x}_{tgt}$ ($S=64$)
Emission lag $S-1=63$
Detection delay $\Delta t = t_{alm} - t^*$
(b) Three-Stage Refitted Permutation Decomposition ($q=0.1\%$)
Realized Held-Out Causal FPR (%, log)
10^1
10^0
10^{-1}
Drift: 57%
Serial: 35%
1.21%
0.58%
0.19%
Drift: 64%
Serial: 29%
1.85%
0.73%
0.22%
Drift: 78%
Serial: 19%
8.72%
2.04%
0.37%
Drift: 92%
Serial: 5%
7.41%
0.67%
0.32%
Design target $q=0.1\%$
TS-JEPA
OpEntropy
TimesNet
TranAD
Stage 1: Chronological (FPR_{chrono})
Stage 2: Block-64 Shuf. + Refit (FPR_{block})
Stage 3: Point i.i.d. Shuf. + Refit (FPR_{point})